

Check slides for yap
 2 code tables

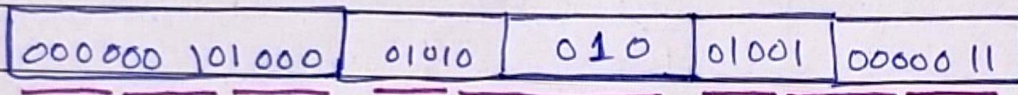
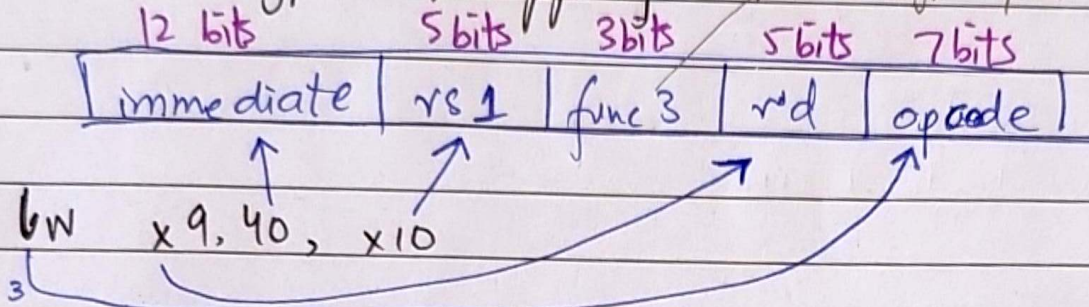
Date _____

* we try to align I & S type with R type

LECTURE-5

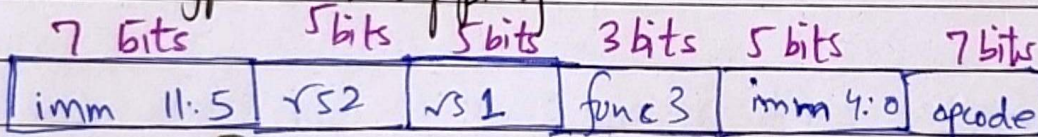
Wednesday
 2/9/26.

⇒ I-Type Mapping: additional bit to define which type of Instrc. is it



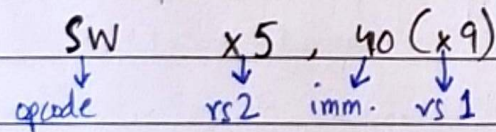
(0 2 8 5 2 4 8 3)₁₆

⇒ S-Type Mapping:



7 MSB of immediate are here

5 MSB of imm are here



⇒ Logical Shifts:

- Shift left logical (SLL): Shift all bits to left and add zero(s) on right.
- Shift left arithmetic (SLLA):

- Shift right logical (SRL): Shift all to right and add zero(s) on left.
- Shift right arithmetic (SRA): Shift all bits to right and put the old MSB to new MSB on left.

1100101 $\xrightarrow{\text{SRA}}$ 1110010

Braching: Conditional branching

`beq x5, x6, 100`

branch if equal : if values of x5 and x6 are equal ;
jump to PC+100 address

⇒ Jump & Link: ~~Unconditional~~ Unconditional Jumps

This will jump chahay jo bhi hojaye; but will return
to the area jahan say jump lagai

Mapping:

① C Code: $A[12] = h + A[9]$

h in x21, base address of A in x22.

RISC-V Code:

`ld x9, 12(x22)`

`add x9, x21, x9`

`sd x9, 96(x22)`

→ $12 \times 8 = 96$

Since it's a 'ld' not 'lw'

∴ we've double word of 8 bytes

So we'll multiply the offset i.e: '9'
with 8 so $9 \times 8 = 72$

② C-Code: $\text{if } (i == j) \{ f = g + h \}$
 $\text{else } \{ f = g - h \}$

f, g ... in x19, x20

RISC-V Code:

`bne x22, x23, Else`

`add x19, x20, x21`

`beq 0x, 0x, Exit`

Else: `sub x19, x20, x21`

Exit: ...